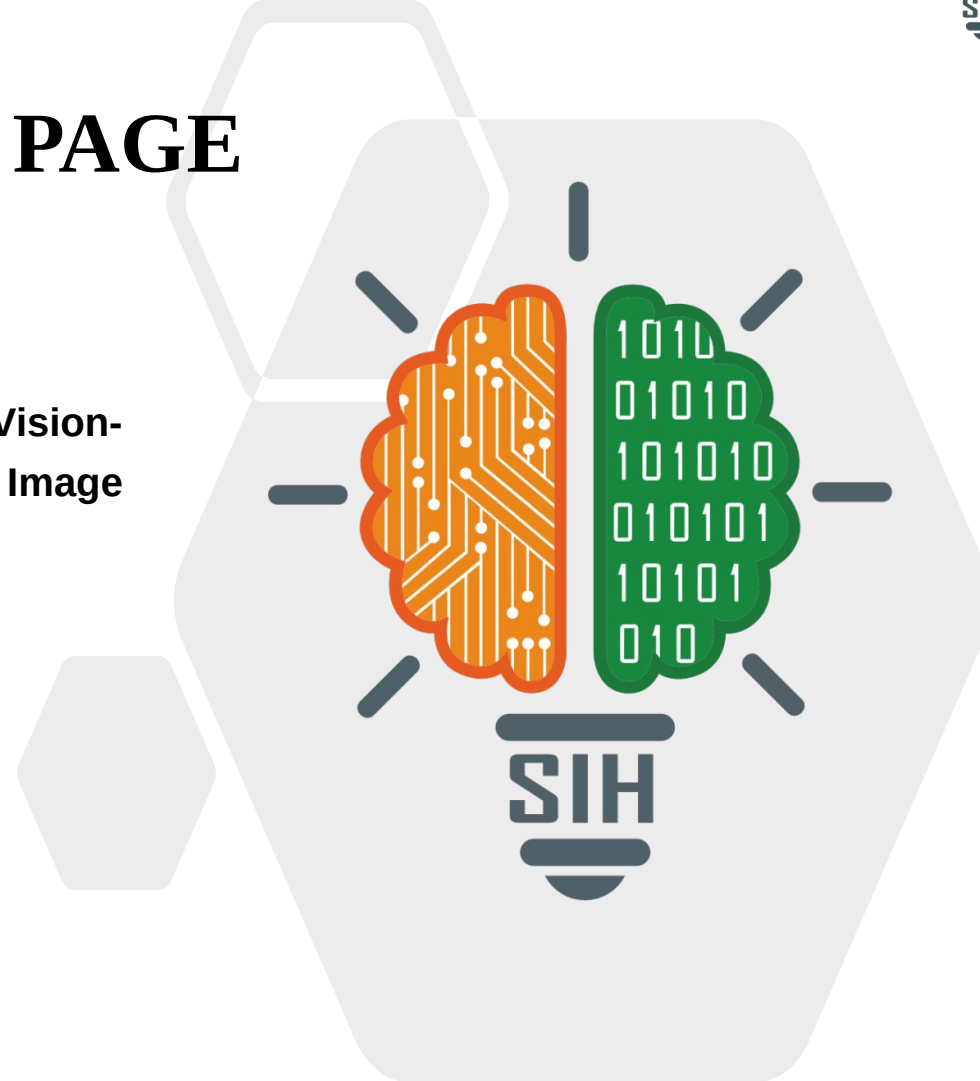




TITLE PAGE

- Problem Statement ID – 26167
- Problem Statement Title- SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- Theme- Space Technology
- PS Category- Software
- Team ID- (as registered on portal)
- Team Name (Registered on portal)- Lunar Circle



SatQuery AI

❖ Proposed Solution (Describe your Idea/Solution/Prototype)



What it is

An agentic vision-language assistant — upload GeoTIFF satellite imagery, ask in plain language, get an evidence-grounded answer with masks/boxes, confidence and an auditable execution trace.



Covers the full prescribed scope

Single optical/SAR image · co-registered optical-SAR pair · bi-temporal pair — across VQA, captioning, grounding, change-VQA and optical-SAR fusion.



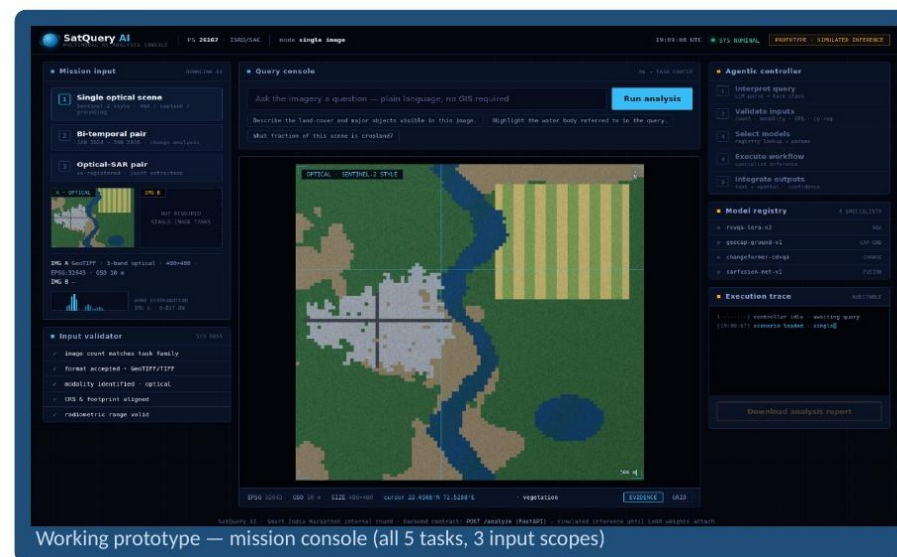
How it addresses the problem

Replaces fragmented single-task RS tools: a controller validates every input and routes each query to fine-tuned specialist models — no GIS expertise needed.



Innovation & uniqueness

Agentic orchestration (not one generic VLM) · LoRA adaptation on BigEarthNet.txt (464,044 co-registered S1 SAR + S2 pairs, 9.6M annotations) · rejects incompatible inputs · every answer ships visual evidence.



LIVE NOW — TRY IT
[gangadhar017.github.io/Lunar_circle](https://github.com/gangadhar017/Lunar_circle)

Technologies to be used



Methodology & process — agentic pipeline (working prototype)



SCAN —
LIVE
PIPELINE
DEMO

Analysis of feasibility

✓ 100% open data

BigEarthNet.txt, VRSBench, RSVQA, CDVQA — training & evaluation start on day one.

✓ Strong open baselines

GeoChat, RemoteCLIP and change models exist — we adapt, not reinvent.

✓ Fits a free GPU

LoRA fine-tunes <1% of parameters — trains on a single Colab/Kaggle T4.

✓ Modular teamwork

Each specialist is built & tested independently, then registered.

✓ Already deployed

Frontend live on GitHub Pages; Dockerised FastAPI backend one click from Render.

Potential challenges & risks



Limited GPU compute for VLM training



LoRA/PEFT + 4-bit quantised inference; frozen backbone, small adapters.



GeoTIFF & multi-band sensor handling



GDAL preprocessing: band selection, normalisation, tiling, CRS checks.



Domain gap — Cartosat-2S / RISAT eval



Multisensor S1+S2 training + SAR-style speckle & resolution augmentation.



VLM hallucination on unfamiliar scenes



Grounded masks/boxes, confidence thresholds, strict input validator.



Live orchestration failure



Deterministic routing + full trace + graceful fallback to single-image VQA.

Potential impact on the target audience — no GIS training required



Agriculture

crop & land-cover monitoring each season by simple queries



Disaster response

SAR sees through clouds — flood answers day or night



Urban planning

“has built-up area grown?” answered with change maps



Forests & ecology

deforestation & vegetation change tracked over time



Water resources

water bodies grounded across optical + SAR



Infrastructure

assets mapped & monitored from multimodal imagery

Benefits of the solution



SOCIAL

Democratises ISRO Earth-observation data — district officials, farmers' bodies and NGOs question imagery in plain language.



ECONOMIC

Hours of manual multi-tool GIS analysis → seconds per query; one platform replaces many single-task systems.



ENVIRONMENTAL

Continuous monitoring of forests, water and sprawl; SAR works through clouds, day & night — critical in floods and cyclones.



STRATEGIC

Multiplies the value of Cartosat & RISAT missions; auditable, evidence-grounded AI fit for official governance use.

Datasets (as prescribed by PS 26167)



BigEarthNet.txt — training

arXiv:2603.29630 · txt.bigearth.net — 464,044 co-registered Sentinel-1 SAR + Sentinel-2 pairs, 9.6M annotations (captions · VQA · referring-expression detection)



VRSBench — evaluation

github.com/lx709/VRSBench — captioning, grounding & VQA



RSVQA — evaluation

rsvqa.sylvainlobry.com — remote-sensing VQA (LR/HR)



CDVQA — evaluation

github.com/YZHJessica/CDVQA — change-based VQA

Key research we build on



GeoChat (CVPR 2024)

grounded VLM for remote sensing — adaptation base



RemoteCLIP (2024)

RS vision-language representation learning



RSVQA (IEEE TGRS 2020) · CDVQA (IEEE TGRS 2022)

task formulations for RS-VQA and change-VQA



LoRA (ICLR 2022)

parameter-efficient fine-tuning under low compute

Evaluation-ready: prescribed public test splits with normalised scores (accuracy · BLEU/CIDEr · IoU); multisensor training targets the ISRO/SAC Cartosat-2S + RISAT hidden set.



LIVE PROTOTYPE [gangadhar017.github.io/Lunar_circle](https://github.com/gangadhar017/Lunar_circle) **SOURCE + BACKEND** github.com/Gangadhar017/Lunar_circle